

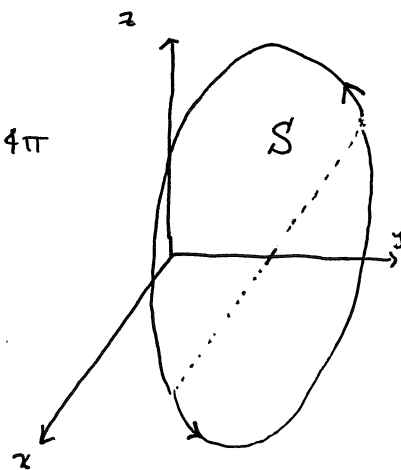
Problem 1: Let C be the boundary of the disc $x^2 + z^2 \leq 4$ in the $y = 1$ plane with positive orientation.

Compute $\oint_C \vec{F} \cdot d\vec{r}$ for the vector fields: (a) $\vec{F} = \left[\frac{M}{5z+2xz}, \frac{N}{1}, \frac{P}{x^2+y} \right]$ and (b) $\vec{F} = \left[\frac{M}{5zy}, \frac{N}{x^2}, \frac{P}{y} \right]$

$$(a) \quad \text{curl } \vec{F} = [1-0, 5+2x-2x, 0-0] = [1, 5, 0]$$

Since $\text{curl } \vec{F} \cdot \vec{n} = 5$ (constant) and $\text{Area}(S) = 4\pi$
then by Stokes' Thm

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S 5 dA = 5 \text{Area}(S) = 20\pi$$



$y=1$ plane
 $\vec{n} = [0, 1, 0]$

$\text{Area}(S) = 4\pi$

$$(b) \quad \text{curl } \vec{F} = [1-0, 5y \quad -0, 2x-5z] \\ = [1, 5y, 2x-5z]$$

Since $\text{curl } \vec{F}$ is non-constant, let's parameterize S :

$$\begin{aligned} x &= r \cos \theta \\ y &= 1 \\ z &= r \sin \theta \end{aligned} \quad \xrightarrow[u=\theta]{u=r} \quad \vec{r}(u,v) = [u \cos v, 1, u \sin v] \quad \left. \begin{aligned} 0 \leq u \leq 2 \\ 0 \leq v \leq 2\pi \end{aligned} \right\} R$$

$$\begin{aligned} \Rightarrow \quad \vec{r}_u &= [\cos v, 0, \sin v] \\ \vec{r}_v &= [-u \sin v, 0, u \cos v] \end{aligned} \quad \Rightarrow \quad \vec{N} = [0, u \cos^2 v - u \sin^2 v, 0] \\ = [0, -u, 0] \quad \leftarrow \text{Note: } \vec{N} \text{ is in the wrong direction}$$

$$\text{curl } (\vec{F}(\vec{r})) = [1, 5, 2u \cos v - 5u \sin v]$$

$$\Rightarrow \text{curl } (\vec{F}(\vec{r})) \cdot (-\vec{N}) = 5u$$

so we should use $-\vec{N}$ for our normal.

$$\begin{aligned} \Rightarrow \oint_C \vec{F} \cdot d\vec{r} &= \iint_R 5u \, du \, dv = \int_0^{2\pi} \int_0^2 5u \, du \, dv \\ &= 2\pi \left(\frac{5}{2} u^2 \Big|_0^2 \right) \\ &= 2\pi \left(\frac{5}{2} (4) \right) = 20\pi \end{aligned}$$